

Weekly Assignment: Direct Proof & Counterexample (Epp 4.1–4.4)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 4.1 #2). Assume that c is a particular integer. Justify your answers by using the definitions of even and odd.

- a. Is $-6c$ an even integer?
- b. Is $8c + 5$ an odd integer?
- c. Is $(c^2 + 1) - (c^2 - 1) - 2$ an even integer?

Problem 2 (Epp 4.1 #9). Prove the following statement:

There is a real number x such that $x > 1$ and $2^x > x^{10}$.

Hint: If you're stuck, think about taking the log of both sides.

Problem 3 (Epp 4.1 #13). Consider the following statement:

For every integer n , if n is odd then $\frac{n^2 - 1}{2}$ is odd.

- (a) Write a negation for this statement.
- (b) Use a counterexample to disprove the statement. Explain how the counterexample actually shows that the statement is false.

Problem 4 (Epp 4.2 #5). Prove the following statement directly from the definitions of even and odd. Follow the directions for writing proofs given in Section 4.2.

If a and b are any odd integers, then $a^2 + b^2$ is even.

Problem 5 (Epp 4.3 #10). Assume that m and n are both integers and that $n \neq 0$. Explain why $\frac{5m^2 + 2n}{4n}$ must be a rational number.

Problem 6 (Epp 4.4 #17). Prove the following statement directly from the definition of divisibility:

For all integers a , b , c , and d , if $a \mid c$ and $b \mid d$ then $ab \mid cd$.

Problem 7 (Proof: Rotations as a Group). Consider an object that can be rotated around its vertical axis, either clockwise or counterclockwise. We can think of all possible rotations as a set G , with the binary operation being *composition*: perform the first rotation, then perform the second. (One convenient representation is to identify each rotation with its angle in degrees, so that G is the set of all real numbers modulo 360.)

Prove that this set of rotations forms a **group** under composition. Your proof should address each of the following parts:

- (a) **Identity.** Define the identity element of this group and explain why composing any rotation with it leaves the rotation unchanged.
- (b) **Inverses.** Define the inverse of an arbitrary rotation and explain why composing a rotation with its inverse yields the identity.
- (c) **Group axioms.** Argue that the remaining group axioms hold:
- *Closure:* composing any two rotations produces another rotation in G .
 - *Associativity:* for any three rotations r_1, r_2, r_3 , $(r_1 \circ r_2) \circ r_3 = r_1 \circ (r_2 \circ r_3)$.

You may represent rotations by their degree measure (or radians, or however you like). We are looking for a solid attempt at making the argument rather than any single particular solution—this is practice at doing mathematics like a mathematician!